

Stage 9 End of unit 15

The tests and mark schemes have been written by the authors. These may not fully reflect the approach of Cambridge Assessment International Education.

Answers

1 a

Age, a (years)	Frequency	Midpoint
$20 \leq a < 30$	6	25
$30 \leq a < 40$	12	35
$40 \leq a < 50$	14	45
$50 \leq a < 60$	8	55

[1]

b 40

[1]

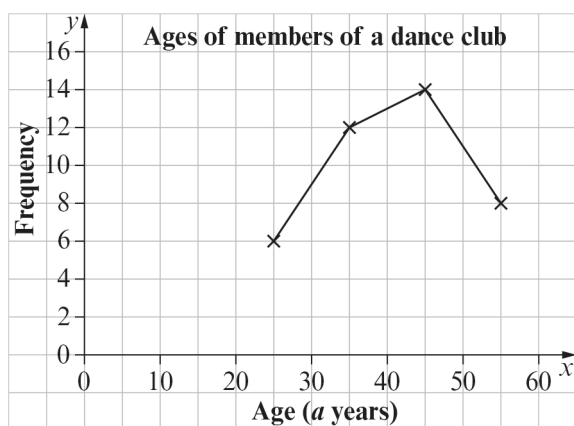
c $\frac{4}{5}$

[1]

d False. Learner's own explanation. For example: It is impossible to tell how old the youngest person is. All you know is that the youngest person is in the age group $20 \leq a < 30$. You do not know the exact ages of the members.

[2]

e



[4]

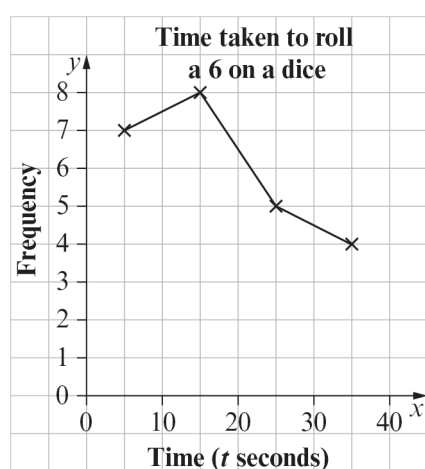
2 a Learner's choice of class intervals. For example:

[4]

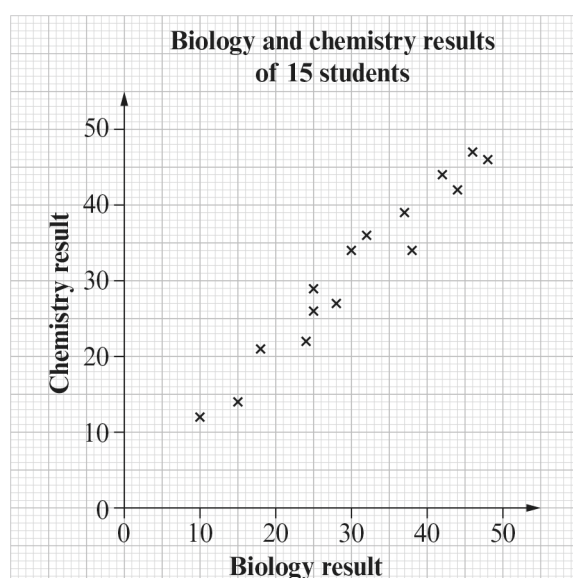
Time, t seconds	Tally	Frequency	Midpoint
$0 < t \leq 10$	II	7	5
$10 < t \leq 20$	III	8	15
$20 < t \leq 30$		5	25
$30 < t \leq 40$	IIII	4	35

b Correct frequency polygon for their class intervals. For example:

[4]



3 a



[4]

- b** Positive [1]
- c** Learner's own line of best fit. [1]
- d** Correct answer for learner's line of best fit. [1]

4 a

Newgale beach	Solva car park
9 2	2 3 4 4
9 8 7 4 3 3	3 5 6 8 9
9 5 2 2 2 1 1 4	0 1
0 5	2 5 8

Key: For Newgale beach, 9 | 2 means 29 burgers

For Solva car park, 2 | 4 means 24 burgers

[4]

b

Location	Mode	Median	Range
Newgale beach	42	41	21
Solva car park	24	37	36

[3]

- c** Learner's own answers. For example:

At Solva car park the number of burgers sold each day was more varied than at Newgale beach. Using the median and mode you can say that on average the number of burgers sold each day was greater at Newgale beach than at Solva car park.

[3]

- d** Learner's own answers. For example:

I agree because the median and mode are both greater at Newgale beach than at Solva car park.

[2]

- 5 a** 40

[1]

b

Station	Modal class interval	Class interval where the median lies	Estimate of mean
Ashton	$0 \leq t < 10$	$10 \leq t < 20$	15.75
Burton	$0 \leq t < 10$	$0 \leq t < 10$	17.5

[4]

- c** Learner's own answers. For example:

The median delay is less at Burton station than at Ashton station. However, the mean delay is less at Ashton station than at Burton station. The modal class interval is the same for both stations.

(Answers must be supported by the data to get the marks)

[2]

- d** Learner's own answers. For example:

I would rather go to Ashton station because the mean delay is less. Also, Ashton station had less than half the number of trains delayed between 30 and 40 minutes compared to Burton station.

[2]

[Total: 45 marks]